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Original Article

Nutritional knowledge, dietary practices, and nutritional status among rural adolescent girls in Bangladesh: a cross-sectional study

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ABSTRACT

Background: Adolescence is a critical phase for physical and cognitive growth. Healthy adolescent girls influence future maternal and child health outcomes, yet malnutrition among adolescents' girls remains a neglected concern in low- and middle-income countries. In Bangladesh, approximately 31% of adolescent girls are underweight and micronutrient deficiencies remain highly prevalent, reflecting persistent nutritional challenges. This study aimed to assess the nutritional knowledge, dietary practices, and nutritional status of adolescent girls in rural Bangladesh and identify associated factors.

Methods: A descriptive cross-sectional study was conducted among 311 adolescent girls aged 10–19 years from a rural school and a college in Patuakhali District, using a purposive sampling technique between February and September 2024. Data were collected using a pre-tested, interviewer-administered questionnaire and anthropometric measurements. Nutritional knowledge and practices were scored dichotomously based on responses to structured questions, with predefined cut-offs used to classify knowledge and practice levels as good or poor, while nutritional status was determined using BMI-for-age percentiles based on WHO standards. Data were analyzed in SPSS v25 using descriptive statistics and Chi-square tests at a significance level of $P < 0.05$.

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Results: Only 22.5% of respondents demonstrated good nutritional knowledge level, and 65.3% had good dietary practices. While 70.4% maintained a healthy BMI, 19.6% were underweight, 7.4% overweight, and 2.6% obese. Significant associations were found between nutritional status and level of education ($P = 0.032$) and nutritional knowledge ($P < 0.001$), dietary practice ($P = 0.002$), and health condition ($P < 0.001$). Poor knowledge on nutrition, limited access to nutritious food, and skipping breakfast were significantly associated factors for undernutrition. Nearly 90% had never undergone nutritional assessment, and 64% reported at least one clinical symptom of nutrition deficiency.

Conclusion: Rural adolescent girls in Bangladesh face a dual burden of malnutrition associated with poor nutrition knowledge, socioeconomic barriers, and inadequate dietary diversity. School-based nutrition education and improved access to affordable nutritious foods are crucial for breaking the intergenerational cycle of malnutrition.

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Introduction

Adolescence is the period between 10 and 19 years of age [1]. Globally, there are approximately 1.2 billion adolescents, about 90% residing in low- and middle-income countries (LMICs) [2]. During this phase, individuals gain up to 50% of their adult weight, achieve 20% of their final height, and develop around half of their adult bone mass [3]. Healthy adolescent girls influences future maternal and child health outcomes [4,5].

Globally, adolescent malnutrition presents in multiple forms [6]. The WHO estimates that nearly one in five adolescents worldwide suffers from some form of malnutrition. In LMICs, inadequate dietary diversity and the predominance of starchy staples are major causes of poor nutrition [7]. Simultaneously, a growing trend of consumption of processed foods and sugary beverages has resulted in a rise in overweight and obesity among young people [8]. The consequences of adolescent malnutrition impair cognitive development, that effects on future economic productivity, and creates an intergenerational cycle of poor health outcomes [9,10].

Despite global advancements in food systems, adolescent nutrition for girls continues to receive insufficient policy and programmatic attention in South Asia [11]. The South East Asia region exhibits the world's highest rates of adolescent undernutrition, driven by poverty, limited access to nutrient-rich foods, and gender inequality [12].

The nutritional landscape in Bangladesh is complicated by both undernutrition and the emerging problem of overweight and obesity [13]. Research indicates that Bangladeshi adolescents' diets are monotonous, often dominated by rice, with inadequate intake of fruits, vegetables, and animal-source foods [14], that contributes to widespread deficiencies in iron, calcium, and vitamins [15]. Studies have shown that more than 60% of schoolgirls in Dhaka consume less than 75% of the recommended essential nutrients [16] that are linked to fatigue, impaired immunity, and adverse effect on maternal and neonatal outcomes [17]. Malnourished adolescent girls are more likely to become malnourished mothers, perpetuating a cycle of undernutrition across generations [18]. Furthermore, low family income, limited parental education, and restricted autonomy of girls significantly influence dietary behaviors in rural Bangladeshi communities [19].

Adolescents with better nutrition knowledge are more likely to make healthier food choices and maintain appropriate body weight [20–22]. Conversely, misinformation and lack of awareness contribute to poor eating habits and nutritional deficiencies [23]. Schools, mass media, and

community programs are potential platforms for promoting nutritional awareness; however, their reach and effectiveness in rural Bangladesh remain underexplored [24]. Hence, understanding the current level of nutrition knowledge and dietary practices among adolescent girls is essential for designing evidence-based interventions. Patuakhali District represents a typical rural coastal setting in Bangladesh characterized by limited healthcare access, socioeconomic vulnerabilities, and food insecurity, making it an appropriate context for examining adolescent nutritional challenges.

Despite substantial progress in child nutrition research, limited data exist on the effectiveness of nutrition information dissemination among school-going adolescents in rural areas [20,21]. Most existing studies focus on early childhood malnutrition, leaving a significant gap in understanding the dietary habits, nutritional knowledge, and health status of adolescent girls, especially in rural contexts [25]. However, there is a lack of comprehensive evidence integrating nutritional knowledge, dietary practices, and anthropometric nutritional status among rural adolescent girls in Bangladesh, particularly at the community and school level.

Therefore, this study aimed to: (1) assess the level of nutritional knowledge among rural adolescent girls, (2) evaluate their dietary practices, (3) determine their nutritional status using BMI-for-age, and (4) examine associations between socio-demographic and behavioral factors with nutritional status.

Methodology

Study Design and Settings: This was a descriptive cross-sectional study to assess the nutritional knowledge level and dietary practices of rural adolescent girls of a purposively selected school and a college under Kanakdia union of Bauphal Upazila, Patuakhali, Bangladesh. This design was appropriate for estimating prevalence and examining associations between variables within a defined population at a single point in time. The area is predominantly agrarian, with moderate poverty levels, limited dietary diversity, and restricted access to specialized nutrition services, which may influence adolescent nutritional outcomes. Institutions were selected based on accessibility, rural representation, and administrative approval. The investigation was conducted between February and September 2024.

Study Population and Sampling Technique: The study population comprised adolescent girls aged 10–19 years who were enrolled in the selected school and college during the study period. The sample size was determined using Cochran's formula for estimating proportions in large populations, expressed as $n_0 = (Z^2 \times p \times (1 - p))/e^2$, where Z represents the standard normal deviate at a 95% confidence level (1.96), p is the estimated proportion of the population possessing the attribute of interest (assumed as 0.5 to maximize sample size), and e is the margin of error (0.05). This yielded an initial sample of 384 participants. After adjusting for participant availability and scrutinizing the collected information 311 respondents were found valid for analysis. A purposive sampling approach was used to select one secondary school and one college based on accessibility and willingness to participate. All eligible adolescent girls present during data collection days were invited. The response rate was approximately 81% (311/384).

Inclusion and Exclusion Criteria: Adolescent girls aged 10–19 years who were enrolled in the selected institutions, available during the study period, and willing to participate were included. Students those who declined participation, or those were absent on the day of data collection were excluded. Exclusion criteria additionally included pregnant adolescents and those with known chronic illnesses that could influence nutritional status.

Data Collection Instruments and Procedures: A pre-tested, interviewer-administered semi-structured questionnaire was used for data collection. The instrument included sections on socio-demographic characteristics, nutritional knowledge, dietary practices, physical activity, and anthropometric measurements. Prior to implementation, the questionnaire was pilot-tested among adolescents from a non-sampled school to assess clarity and cultural appropriateness. Minor wording adjustments were made following pilot testing; no major structural changes were required. The questionnaire was adapted from previously validated tools used in similar studies and demonstrated acceptable internal consistency (Cronbach's alpha = 0.78).

The knowledge on nutritional level encompassed understanding of nutritious food, awareness of micronutrient sources (vitamin A, vitamin C, and dietary iron), knowledge of anemia, familiarity with the health functions of nutrients, and dietary diversity concepts. Dietary practice items captured routine behaviors such as meal frequency, breakfast consumption, fruit and vegetable intake, snacking patterns, and hygiene-related practices. The principal investigator collected data by face-to-face by interview in a private space within the school premises to ensure confidentiality and participant comfort.

Study Variables and Operational Definitions: The primary outcome variable was nutritional status, measured by Body Mass Index (BMI) according to World Health Organization (WHO) reference standards for children and adolescents. Independent variables encompassed socio-demographic attributes (age, religion, education, and family income), behavioral characteristics (nutritional knowledge and dietary practices), and lifestyle indicators (physical activity and sedentary behavior).

In this study, "Knowledge" refers to the awareness and understanding of nutrition and its importance, assessed through a set of structured questions scored dichotomously as correct or incorrect [26]. "Practice" indicates the application of nutrition-related behaviors, such as healthy eating and physical activity, evaluated through self-reported responses. Nutritional status was measured using Body Mass Index (BMI), according to WHO standards [27]. Adolescence was defined as the age range of 10–19 years, consistent with WHO criteria. Moderate physical activity was defined as brisk walking, cycling, or active play; vigorous activity involved activities like running or competitive sports, consistent with WHO adolescent activity guidelines.

Nutritional knowledge was scored by assigning one point for each correct response and zero for incorrect ones, with a cumulative score of $\geq$5 indicating good knowledge (total knowledge score ranged from 0 to 7). Dietary practices were assessed using a structured food frequency approach combined with behavior-based questions, such as-breakfast habits, meal frequency, fruit and vegetable intake, snacking behaviors, and hygiene practices. Each healthy behavior contributed one point, and a total score of $\geq$3 denoted good dietary practice. Additional practice variables incorporated into the analysis were screen use during meals, post-meal behavior, and use of extra salt.

Data Quality Control: The questionnaire was refined through pre-testing, and field data were reviewed daily for completeness and consistency. Anthropometric measurements were taken following standardized procedures. Measurements were taken twice and averaged to minimize error. Weight was measured with a calibrated digital scale to the nearest 0.1 kg, and height was measured with a stadiometer to the nearest 0.1 cm. Responses were double-entered and cross-checked in SPSS before analysis.

Data Analysis: Data were analyzed using Statistical Package for the Social Sciences (SPSS) version 25. Descriptive statistics, were used to summarize socio-demographic, behavioral, and nutritional variables. Associations between nutritional status with nutritional knowledge, dietary practice, and socio-economic characteristics were examined using the Chi-square test. Statistical significance was determined at a *P*-value threshold of <0.05. Due to the exploratory nature and sample size constraints, multivariable regression was not performed; however, future studies should consider adjusted analyses.

Results

Table 1 presents the socio-demographic characteristics of the respondents. The mean age of participants was 14.84 years (SD = 2.13), with a slight predominance of older adolescents (54.1% aged 15–19 years). The majority were enrolled in secondary-level education (76.9%), and nearly half of the respondents (48.6%) came from households with a monthly income below 20,000 BDT, indicating a predominantly lower socioeconomic background. Most participants were Muslim (87.8%), and the typical family size was three to five members (70.8%).

Table 2, reveals that only 40.2% of respondents were aware of nutritious food and its importance, 58.2% could identify at least three sources of vitamin C, only 34.1% could name three dietary sources of iron, 23.2% of the respondents reported involvement in nutrition education or training, and 21.5% was familiar with anaemia. Awareness of anemia was also poor, with only 21.5% familiar with the condition. Knowledge regarding the role of vitamin A in vision and immune health was limited (30.5%).

Table 1
Distribution of respondents according to socio-demographic characteristics (n = 311)

Variables	Categories	Frequency (N)	Percentage (%)
Age (years)	10–14	143	45.9
	15–19	168	54.1
	Mean = 14.84, (SD = 2.13)		
Educational level	Class 6-10	239	76.9
	Class 11-12	72	23.1
Monthly family income (BDT)	< 20,000	151	48.6
	20,000–30,000	124	39.8
	> 30,000	36	11.6
Religion	Muslim	273	87.8
	Hindu	38	12.2
Number of family members	3–5	220	70.8
	6–8	81	26.0
	9+	10	3.2

Table 2
Assessment of nutritional knowledge level of the participants (n = 311)

Variables	Yes n (%)	No n (%)
Aware about nutritious food and its importance	125 (40.2)	186 (59.8)
Have knowledge of at least three sources of vitamin C	181 (58.2)	130 (41.8)
Know at least three sources of dietary iron	106 (34.1)	205 (65.9)
Received nutrition education or participated in nutrition-related training	72 (23.2)	239 (76.8)
Have knowledge about anemia	67 (21.5)	244 (78.5)
Know about the benefits of vitamin A for vision and immunity	95 (30.5)	216 (69.5)
Aware of the importance of consuming a variety of colorful fruits and vegetables	199 (64.0)	112 (36.0)

Encouragingly, 64% of participants recognized the importance of consuming a variety of colorful fruits and vegetables, for dietary diversity.

Figure 1 shows that only 22.5% of participants demonstrated good nutritional knowledge (score ≥5), while a substantial majority (77.5%) exhibited poor knowledge, indicating a major gap in nutrition awareness.

Table 3 summarizes the distribution of respondents by physical activity, sedentary behavior, and availability of nutritional support. Less than half of the participants (42.0%) reported the availability of nutrition-related support systems, while a considerable proportion remained uncertain (34.0%). Most respondents reported a fair appetite (80.1%). Regarding lifestyle behaviors, a majority (73.0%) met the criteria for adequate physical activity; however, sedentary behavior was substantial, with 32.5% spending 4–6 hours and 26.7% spending ≥6 hours per day in sedentary activities. Additionally, 60.1% of respondents reported having a food allergy or intolerance, highlighting potential health-related concerns in this population.

Table 4 indicates that 65.3% of participants demonstrated good dietary practice scores, while 34.7% had poor practices. A large majority (73.3%) reported challenges in accessing affordable and nutritious food. Most respondents practiced regular handwashing before meals, with 75.0% reporting always doing so. In terms of food intake patterns, 68.5% consumed three meals per day, and 71.1% reported consuming two snacks daily.

Post-meal behavior showed that 52.7% tended to lay down after meals, whereas only 13.2% engaged in walking. Skipping breakfast was highly prevalent (78.8%). Distracted eating was reported by 34.1% of respondents who watched TV or used mobile phones during meals. While the majority (61.1%) avoided extra salt consumption, 38.9% continued to consume extra salt.

Table 5 shows that rice was universally consumed daily by 100% of respondents. Vegetables (46.9%) and potatoes (38.9%) also showed high daily intake. Fish consumption was 36.7% eating it 5–6 times a week. Moderate intake patterns were evident for eggs, pulses, and chicken, with many consuming them several times per week, indicating dietary diversity and access to protein-rich foods.

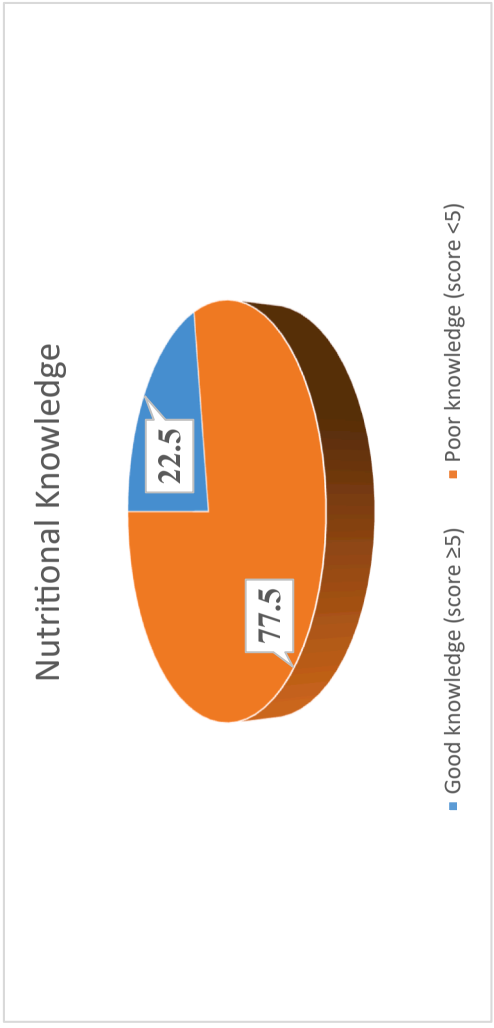


Figure 1. Distribution of the respondents according to level of nutritional knowledge (n = 311).

Table 3
Distribution of respondents by physical activity, sedentary behavior, and availability of nutritional support (n = 311)

Parameter	Category/response	Frequency (n)	Percentage (%)
Availability of nutrition-related support systems	Available	131	42.0
	Not available	75	24.0
	Don't know/uncertain	105	34.0
Appetite	Good	36	11.6
	Fair	249	80.1
	Poor	26	8.4
Frequency of moderate to vigorous physical activity per week	No/once weekly	101	32.5
	2-3 times	125	40.2
	4-5 times	46	14.8
	>5 times	39	12.5
Overall physical activity level	Good (≥3 times/week)	227	73.0
	Poor (<3 times/week)	84	27.0
Sedentary behavior in hours per day	0–2 hours	50	16.1
	2–4 hours	77	24.8
	4–6 hours	101	32.5
	≥6 hours	83	26.7
Food allergy/intolerance	Yes	187	60.1
	No	124	39.9

Table 4
Distribution of respondents by dietary practice score and food intake (n = 311)

Parameter	Category/response	Frequency (n)	Percentage (%)
Dietary practice score	Good practice (score ≥3)	108	65.3
	Poor practice (score <3)	203	34.7
Challenges in accessing affordable & nutritious food	Yes	228	73.3
	No	83	26.7
Handwashing practices (before meals)	Always	233	75.0
	Sometimes	78	25.0
Number of meals per day	2 meals	89	28.6
	3 meals	213	68.5
	4 meals	9	2.9
Number of snacks per day	None (0)	55	17.7
	2 snacks	221	71.1
	3 snacks	28	9.0
	3+ snacks	7	2.3
Post-meal behavior	Lay down	164	52.7
	Sit	106	34.1
	Walk	41	13.2
Skipping breakfast	Yes	245	78.8
	No	66	21.2
Watching TV/mobile during meals	Yes	106	34.1
	No	205	65.9
Consuming extra salt	Yes	121	38.9
	No	190	61.1

In contrast, beef and fast food items were least consumed, with 53.1% and 41.2% of respondents, respectively. Milk and yogurt had relatively low intake frequencies, with nearly one-third (28.9%) reporting non-consumption. Seasonal fruits were mostly consumed 1–2 days weekly (44.7%), indicating moderate access to fresh produce. Tea or coffee was consumed daily by 30.2%, showing its popularity as a common beverage. Conversely, soft drinks, snacks, sweets, and fast food were infrequently consumed.

Table 6 revealed that 70.4% maintained a healthy BMI, 19.6% underweight, 7.4% overweight, and 2.6% obese individuals. Alarmingly, 90% of respondents had never undergone a nutritional assessment, indicating limited awareness or access to nutritional evaluation services. Frequent fatigue (49.2%) and hair loss (45.3%) were the most common symptoms of nutritional deficiency, followed by

Table 5
Weekly intake of selected food items by respondents

Food item	7 days (%)	5–6 days (%)	3–4 days (%)	1–2 days (%)	0 days (%)
Rice	100.0	0.0	0.0	0.0	0.0
Milk/Yogurt	12.5	10.0	16.1	32.5	28.9
Chicken	5.5	11.3	26.7	51.1	5.5
Beef	0.0	1.0	3.2	42.8	53.1
Fish	23.8	36.7	23.5	13.5	2.6
Eggs	21.2	23.8	25.7	24.1	5.1
Vegetables	46.9	27.3	8.7	13.8	3.2
Potatoes	38.9	28.0	15.1	12.9	5.1
Pulses	37.6	17.7	23.2	16.1	5.5
Seasonal Fruits	11.3	11.3	20.6	44.7	12.2
Tea/Coffee	30.2	8.4	9.0	23.8	28.6
Soft Drinks	8.0	6.4	13.2	39.5	32.8
Snacks	17.7	23.2	17.4	29.3	12.5
Sweets	3.9	7.1	14.1	45.0	29.9
Fast Food Items	5.8	9.3	9.3	34.4	41.2

Table 6
Nutritional and health condition of respondents based on BMI and nutritional deficiency symptoms

Parameter	Category	Frequency (n)	Percentage (%)
Nutritional status according to BMI range	Healthy weight (5th - < 85th percentile)	219	70.4
	Underweight (<5th percentile)	61	19.6
	Overweight (85th - <95th percentile)	23	7.4
	Obese (≥95th percentile)	8	2.6
Nutritional assessment history	Ever assessed	31	10.0
	Never assessed	280	90.0
Symptoms related to nutritional deficiencies[a]	Frequent fatigue	153	49.2
	Hair loss	141	45.3
	Frequent illness	52	16.7
	Others (easy bruising, muscle weakness, poor night vision, mood changes like irritability)	30	9.6
	Pale skin	24	7.7
	Brittle nails	20	6.4
Overall health condition based on symptoms	Good health (no symptom)	112	36.0
	Poor health (≥1 symptom)	199	64.0

[a] Multiple responses were allowed in symptom reporting.

frequent illness (16.7%), pale skin (7.7%), brittle nails (6.4%), and other minor symptoms (9.6%). Overall, 64% of participants were classified as having poor health based on the presence of one or more deficiency-related symptoms.

Table 7 shows that the participants with higher education ($P = 0.032$), good nutritional knowledge ($P < 0.001$), and better dietary practices ($P = 0.002$) were more likely to have good nutritional status. Similarly, good health based on symptoms showed a highly significant relationship with nutritional well-being ($P < 0.001$). In contrast, age, family income, and information about dietary history did not show significant associations.

When considering underweight status, challenges in accessing affordable and nutritious food ($P < 0.001$), poor nutritional knowledge ($P = 0.002$), inadequate dietary practices ($P = 0.002$), and poor health status ($P < 0.001$) were strongly linked to undernutrition. Furthermore, overweight status demonstrated significant relationships with fast food consumption ($P = 0.013$) and the presence of symptoms ($P = 0.020$).

Discussion

This study explored the nutritional knowledge, dietary practice, food habits, and nutritional status of adolescent girls in a rural area of Patuakhali district of Bangladesh. The findings address the study

Table 7
Factors associated with nutritional status among study participants (n = 311)

Variables	Categories	Nutritional status-Good (n)	Nutritional status-Poor (n)	Total	χ^2	*P*-value
Age	10–14 years	97	46	143	0.850	0.357
	15–19 years	122	46	168		
Education	Class 6-10	161	78	239	4.622	0.032
	HSC	58	14	72		
Monthly family income	< 20,000 BDT	113	47	160	0.007	0.934
	>20, 000 BDT	106	45	151		
Nutritional knowledge level	Good	62	8	70	14.290	< 0.001
	Poor	157	84	241		
Nutritious food intake practice	Good	64	44	108	9.890	0.002
	Poor	155	48	203		
Information about dietary history	Good	101	43	144	0.010	0.920
	Poor	118	49	167		
Symptoms according to nutrition related health condition	Good	95	17	112	17.432	< 0.001
	Poor	124	75	199		
Challenges in accessing nutritious food	Yes	58	167	225	17.398	< 0.001
	No	4	82	86		
Underweight and nutritional knowledge	Good	5	65	70	9.262	0.002
	Poor	57	184	241		
Underweight and nutritious food intake practice	Good	32	76	108	9.741	0.002
	Poor	30	173	203		
Underweight and health condition	Good	10	102	112	13.286	< 0.001
	Poor	52	147	199		
Overweight and fast food consumption	Yes	64	13	77	6.149	0.013
	No	217	17	234		
Overweight and symptoms present	Good	107	5	112	5.393	0.020
	Poor	174	25	199		

objectives by revealing clear associations between nutritional knowledge, dietary practices, and overall nutritional status. These findings align with the global understanding of adolescent nutritional behaviors [1,4]. Across low- and middle-income countries (LMICs), adolescent girls face a “double burden” of malnutrition, experiencing both undernutrition and increasing trends of overweight due to dietary transitions [6,8]. The results of this study reaffirm that Bangladesh continues to mirror this global trend, with inadequate dietary diversity and nutrition awareness being prominent concerns.

The observed gaps in nutrition knowledge and practices among adolescent girls in rural Bangladesh are consistent with prior research showing that most adolescents lack awareness about essential nutrients and balanced diets [16,14,24]. Limited access to nutrition education in schools and communities remains a significant barrier [21,24]. Similar findings have been reported in Bangladesh [16,24] and India, where poor dietary diversity and limited nutrition knowledge are prevalent among adolescents. Cultural norms and socioeconomic constraints often shape adolescents' food choices, leading to a reliance on carbohydrate-dense, low-protein diets [12,19]. Studies in India and Nepal showed that adolescents in rural areas frequently consume monotonous diets dominated by rice and low intake of fruits, vegetables, and animal-source foods, resulting in micronutrient deficiencies [17,18]. This study's findings suggest a significant association between insufficient nutrition knowledge and suboptimal dietary behaviors and nutritional outcomes among adolescent girls, although causality cannot be established due to the cross-sectional design.

The findings also highlight that families with lower income were more likely to report challenges in accessing nutritious foods, which is consistent with other Bangladeshi studies linking household poverty with higher rates of food insecurity and poor dietary diversity [19,14]. Marginalized populations in LMICs are disproportionately affected by limited food access and low-quality diets due to

economic inequalities [9,10]. In the Bangladeshi context, these inequalities are compounded by rural infrastructural limitations, fluctuating food prices, and seasonal food insecurity [13,28]. These findings can be explained through the socio-ecological model, where individual knowledge, household resources, and environmental constraints interact to shape dietary behaviors and nutritional outcomes. Consequently, improving adolescent nutrition requires addressing not only individual-level knowledge but also structural and socioeconomic barriers to food availability and affordability.

This study reveals that 78.8% of the respondents skip breakfast. Skipping of breakfast and high consumption of processed or fast foods, are increasingly common among adolescents in both urban and rural Bangladesh [17,15]. The rise of such behaviors reflects broader shifts in food systems and changing lifestyle patterns [8]. Comparable findings from urban Indian and Indonesian adolescents showed that fast food contribute to unhealthy dietary patterns and increased risk of overweight and obesity [6,8]. Approximately 73.3% of respondents in this study reported challenges in accessing affordable nutritious food, while about one-fourth consumed fast food at least 3–4 days per week. This emerging nutritional transition among Bangladeshi adolescents indicates that undernutrition and overnutrition coexist within the same population, signaling a complex public health challenge [6,29].

An additional concern is the low level of physical activity and the prevalence of sedentary behaviors reported among participants. Sedentary lifestyles, driven by increased screen time, academic pressure, and limited outdoor opportunities, have been linked with both poor metabolic health and mental well-being among adolescents worldwide [4,17]. Interventions promoting physical activity through school-based programs have demonstrated that integrating nutrition education with active lifestyle promotion significantly improves adolescents' health outcomes in many LMICs [24,30,31]. Evidence from Bangladesh and other LMICs demonstrates that structured nutrition education, delivered through schools, can substantially enhance knowledge, influence attitudes, and foster healthier dietary behaviors [22,24]. Integrating nutrition into the school curriculum, combined with practical initiatives such as school gardens and community nutrition clubs, could be instrumental in bridging existing knowledge gaps. Furthermore, targeted awareness campaigns focusing on the benefits of dietary diversity and micronutrient-rich foods can help foster lasting behavioral change.

Limitations: Its cross-sectional design restricts causal inferences between knowledge, practices, and nutritional outcomes. The study also relied on self-reported data, which may be subject to recall or social desirability bias. Moreover, the research was confined to selected schools in one rural subdistrict, limiting generalizability to other regions.

Conclusion

This study identified suboptimal nutritional knowledge, dietary practices, and a dual burden of malnutrition among rural adolescent girls in Bangladesh. Inadequate awareness of essential nutrients, limited dietary diversity, and socioeconomic constraints were found to be associated with variations in nutritional status. Targeted school-based nutrition education, improved access to affordable nutritious foods, and promotion of physical activity are important public health strategies to support adolescent nutritional well-being. Future studies should employ longitudinal designs to better understand temporal relationships and include biochemical assessments to capture micronutrient deficiencies more accurately. Broader studies involving diverse rural and urban populations may further clarify regional disparities in adolescent nutrition.

Ethical clearance

Ethical approval for this study was obtained from the Institutional Review Board (IRB) of the Faculty of Health and Life Sciences, Daffodil International University, Dhaka, Bangladesh (Ref: REC-FHLS/DIU/2024/0011). Written institutional approvals were collected before data collection.

Human ethics and consent to participate

Prior to data collection, participants were informed about the purpose, procedures, potential risks, and benefits of the study in local language. Written informed consent was obtained from all

participants, and for minors under 18 years, consent was obtained from school authorities. Confidentiality was strictly maintained and data were used solely for academic purposes. The study adhered to the ethical principles of the Declaration of Helsinki. No experiments on humans and/or the use of human tissue samples were done in this study.

Consent for publication

Not applicable.

Availability of data and materials

The data supporting the conclusions of this article are included within the article.

Clinical trial number

Not applicable.

Author contributions

1. Inum Maria: Conceptualization, data curation, formal analysis, investigation, methodology, visualization, and writing – original draft.
2. Md. Monir Hossain Shimul: Methodology, formal analysis, validation, visualization, writing – original draft, and writing – review & editing.
3. Roman Hossain: Formal analysis, visualization, and writing – review & editing.
4. Oiendrila Baroi: Data curation, validation, writing – review and editing.
5. Salamat Khandker: Methodology, and writing – review & editing.
6. Nadira Mehriban: Methodology, supervision, validation, writing – review and editing.

Declaration of generative AI in scientific writing

The authors did not use generative AI or AI-assisted technologies in the development of this manuscript.

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References

[1] de Onis M. Development of a WHO growth reference for school-aged children and adolescents. Bull World Health Organ 2007;85(9):660–7.
[2] Cappa C, Wardlaw T, Langevin-Falcon C, Diers J. Progress for children: a report card on adolescents. Lancet 2012 Jun 23; 379(9834):2323–5. https://doi.org/10.1016/S0140-6736(12)60531-5.

[3] Meier PR, Nickerson HJ, Olson KA, Berg RL, Meyer JA. Prevention of iron deficiency anemia in adolescent and adult pregnancies. Clin Med Res 2003 Jan;1(1):29–36. https://doi.org/10.3121/cmr.1.1.29.
[4] Sawyer SM, Azzopardi PS, Wickremarathne D, Patton GC. The age of adolescence. Lancet Child Adolesc Health 2018 Mar; 2(3):223–8. https://doi.org/10.1016/S2352-4642(18)30022-1.
[5] WHO. Adolescent health and development: WHO Recommendations. Geneva: World Health Organization; 2021.
[6] WHO. The double burden of malnutrition. Policy Brief; 2020.
[7] FAO. Dietary diversity and nutrition transition in LMICs. Rome: FAO; 2022.
[8] Popkin BM, Adair LS, Ng SW. Global nutrition transition and the pandemic of obesity in developing countries. Nutr Rev 2012 Jan;70(1):3–21. https://doi.org/10.1111/j.1753-4887.2011.00456.x.
[9] Akter Z, Shimul MMH, Hossain MK, et al. Nutritional vulnerabilities among Rohingya refugee children in Cox's Bazar: a cross-sectional analysis. Discov Soc Sci Health 2025;5:125. https://doi.org/10.1007/s44155-025-00280-8.
[10] Pasha M, Shimul MMH, Hossain R, Khandker S, Khan S, Mehriban N. Nutritional status and determinants of malnutrition among under-five children in a specialized hospital in Bangladesh during the COVID-19 pandemic. Clin Nutr Open Sci 2025;64:192–202. https://doi.org/10.1016/j.nutos.2025.10.008.
[11] UNICEF. The state of the world's children 2021: on My Mind. New York: UNICEF; 2021.
[12] Shimul MMH, Hossain MK, Khandker S. Workforce and Health Care Services for Young Children in Bangladesh. JAMA Netw Open 2025;8(6):e2513807. https://doi.org/10.1001/jamanetworkopen.2025.13807.
[13] Institute of Public Health Nutrition (IPHN). Nutrition in Bangladesh: status and trends. 2020.
[14] Islam MR, Rahman SM, Tarafder C, Rahman MM, Rahman A, Ekström EC. Exploring Rural Adolescents' Dietary Diversity and Its Socioeconomic Correlates: A Cross-Sectional Study from Matlab, Bangladesh. Nutrients 2020 Jul 26;12(8):2230. https://doi.org/10.3390/nu12082230.
[15] Ali Sarkar MR, Easmin Jharna D, Rabbani Akanda MG, et al. Dietary habits and nutritional status of adolescent girls in environmentally vulnerable coastal areas of Bangladesh. J Health Popul Nutr 2026. https://doi.org/10.1186/s41043-026-01285-y.
[16] National Nutrition Services (NNS), Bangladesh. Nutrition Surveillance report. 2022.
[17] Islam MS, Roy C, Ishadi KS, Mithu MMU, Abedin ES, Karim MS, et al. Dietary Pattern and Nutritional Status of School-going Adolescents in Rural Areas of Bangladesh. Nutr Food Sci 2024;12(3). https://doi.org/10.12944/CRNFSJ.12.3.11.
[18] Parajuli J, Prangthip P. Adolescent Nutrition and Health: a Critical Period for Nutritional Intervention to Prevent Long Term Health Consequences. Curr Nutr Rep 2025 Oct 23;14(1):116. https://doi.org/10.1007/s13668-025-00706-4.
[19] Redwan T, Shabur MA. Analyzing the socio-economic factors of adolescent malnutrition in Bangladesh using Best Worst Method. Discov Sustain 2025;6:260. https://doi.org/10.1007/s43621-025-01121-z.
[20] Government of Bangladesh. National nutrition plan (2021–2030).
[21] Directorate General of Health Services (DGHS), Adolescent health Strategy of Bangladesh 2022–2027.
[22] Contento IR. Nutrition education: linking research, theory, and practice. Asia Pac J Clin Nutr.; 2008.
[23] Bello F, Koukou E, Bodjrenou S, Termote C, Azokpota P, Hounkpatin WA. Food and nutrition knowledge, attitudes and practices among children in public primary school with canteens in southern Benin: a case study. BMC Nutr 2024 Mar 4; 10(1):40. https://doi.org/10.1186/s40795-024-00857-7.
[24] Ponnambalam S, Palanisamy S, Singaravelu R, Arambakkam Janardhanan H. Effectiveness of a school-based nutrition education program on waist circumference and dietary behavior among overweight adolescents in Puducherry, India. J Educ Health Promot 2022 Oct 31;11:323. https://doi.org/10.4103/jehp.jehp_413_22.
[25] Mahmudur Rahman AHM. A Review on Child and Maternal Health Status of Bangladesh. CHRISMED J Health Res Jan-Mar 2018;5(1):1–7. https://doi.org/10.4103/cjhr.cjhr_65_17.
[26] Study instrument definition (from questionnaire).
[27] WHO. BMI Classification and health Implications. 2019. Geneva.
[28] M. M. H. Kazal, C. C. Villinueva, Z. Hossain, and T. K. Das, "Food security strategies of the people living in Haor areas: status and prospects".
[29] Kolčić I. Double burden of malnutrition: a silent driver of double burden of disease in low– and middle–income countries. J Glob Health Dec. 2012;2(2):020303. https://doi.org/10.7189/jogh.02.020303.
[30] Union Statistics.pdf. Available, http://203.112.218.65:8008/WebTestApplication/userfiles/Image/National20Reports/Union20Statistics.pdf. [Accessed 22 September 2024].
[31] Khara T, Mates E. Adolescent nutrition. Policy and programming in SUN+ countries. 2015. https://doi.org/10.13140/RG.2.1.4480.2081.